

Batch Code: ANP_D1838

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LAB-1

Questions:

Task 1

1. What is java and its features?
2. Explain JDK [Java Development Kit], JRE [Java Runtime environment], JVM [Java Virtual Machine].

Task 2

1. Write a simple java program to print Hello World and write a comment.

Answers:

TASK 1

1. JAVA:

- Java is a high-level, object-oriented programming language.
- It was developed by James Gosling.
- Developed at Sun Microsystems company.
- Initially, in 1991, the project name was Oak.
- Later, in 1995, Oak was renamed as Java.
- It is designed around the core philosophy of “Write Once, Run Anywhere”.
- Java Code can run on any device with a Java Virtual Machine (JVM).

Features:

- Java is easy to learn.
- Java Follows class and Objects.
- Platform Independent.
- Java provides Security features.

2. Java Development Kit (JDK):

- JDK is used to develop, compile and run java programs.
- It includes javac compilers, debugger and documentation tools.
- JDK contains:
 - JRE
 - JVM
 - Development tools

Java Runtime Environment (JRE):

- JRE provides the environment to run Java Applications.
- It includes required files to execute java programs.
- JRE contains:
 - JVM
 - Libraries

Java Virtual Machine (JVM):

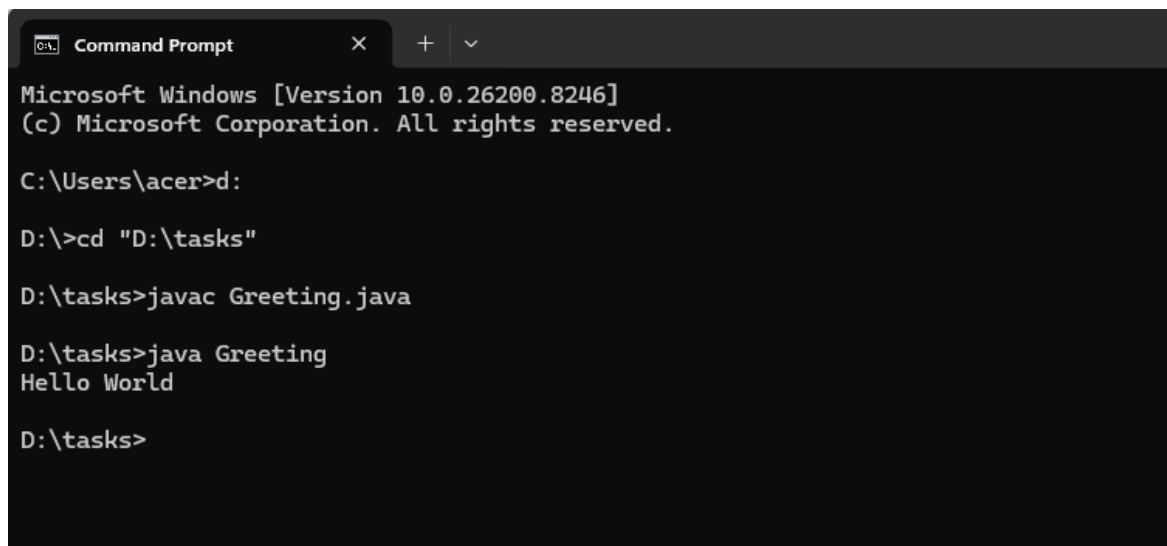
- JVM is responsible for running java Programs.
- JVM converts Bytecode to Machine code.
- It makes Java Platform Independent.
- JVM also handles memory management and garbage collection.

TASK 2

Program:

```
class Greeting {  
    public static void main (String args []) {  
        System.out.println("Hello World");  
    }  
}
```

Output:



```
Microsoft Windows [Version 10.0.26200.8246]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\acer>d:  
  
D:\>cd "D:\tasks"  
  
D:\tasks>javac Greeting.java  
  
D:\tasks>java Greeting  
Hello World  
  
D:\tasks>
```

File name: Greeting.java

To Compile the program: javac Greeting.java

To Run the Program: java Greeting

Comments:

Class	A keyword used to create a class in Java.
Helloworld	Class Name. Name given by the programmer.
Public	Allows the method to be accessed from anywhere.

Static	Allows the method to run without creating an object.
Void	Means the method does not return any value.
Main ()	Main Function. Java program execution starts from here. JVM looks for this method first.
String [] args	An array of Strings used to receive multiple text values from command line.
System	A predefined Java class used for system functions.
Out	Output Object used to display output on the screen.
Println	Method used to print output and move to the next line.
“Hello World”	String Literal. Text value written inside double quotes.
{ }	Curly Braces. Used to show the start and end of a code block.
;	Semicolon. Marks the end of a statement.